Forecasting Next-Day PM2.5 in Dublin Using Mobile Air Quality, Weather, and Traffic Data

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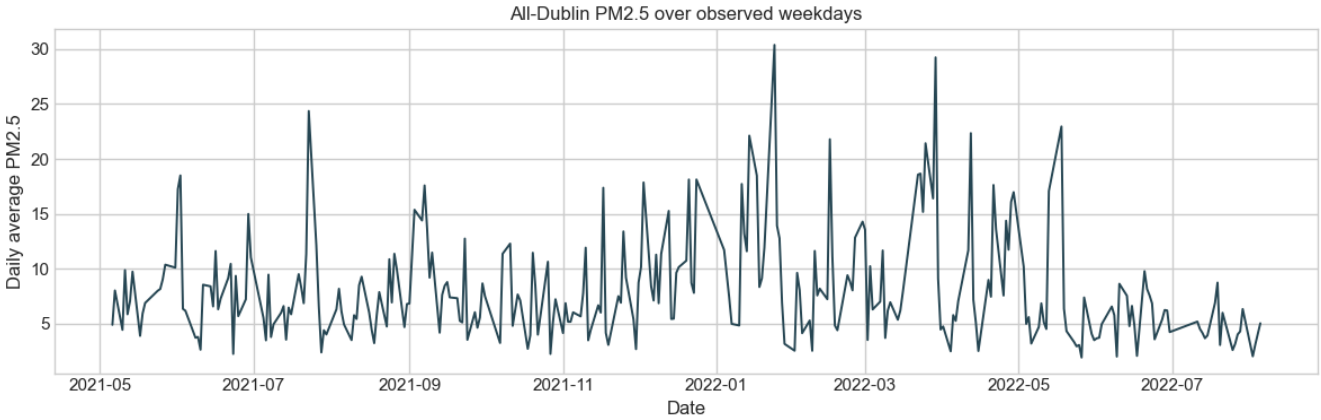
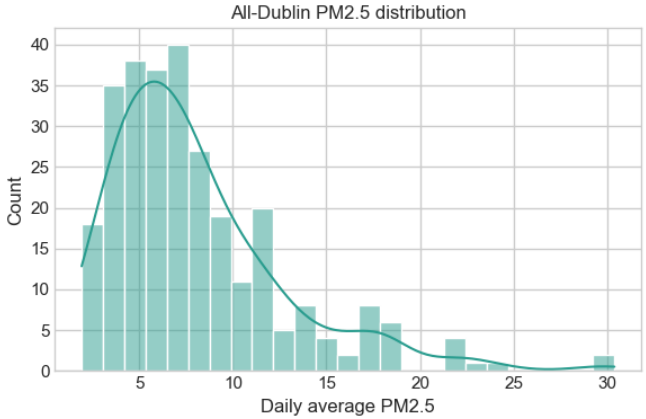
Objective and motivation

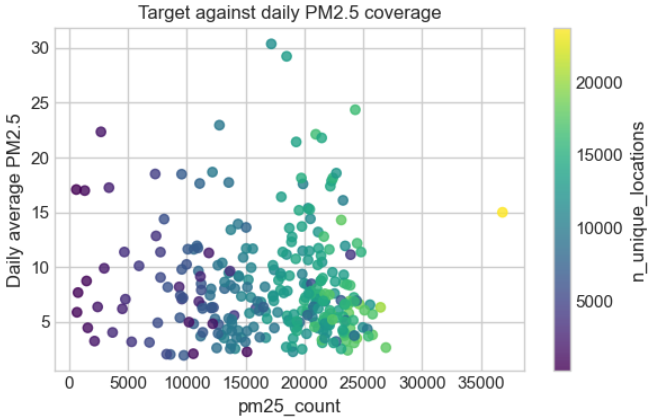
- Research Question: Can next-day air pollution in Dublin be predicted from recent pollution levels, weather and traffic conditions, and what does this imply for DCC's monitoring and intervention strategy?
- Target Variable of Interest: Particulate matter pollution, specifically PM2.5, particles with a size (diameter) of less than 2.5 micrometers
- PM2.5 has been identified as the main contributor to poor health from air pollution as the very small particles can penetrate deep into the respiratory system, contributing to more pollution-related deaths than than ozone (Our World in Data 2025)
- We aimed to create an accurate forecasting model that can predict next day PM2.5 levels, to support Dublin City Council in their monitoring and prevention of air pollution (DCC Controlling Air Pollution)

Data Collected

- Google Air View Pollution Data:
- Metrological Data:
- Traffic Data:
- Critical decision , the weekday , lagss (lag 1)

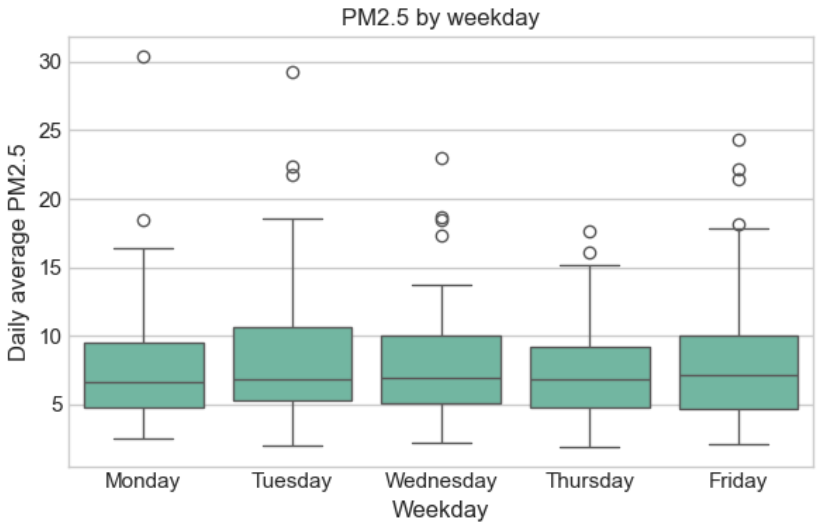


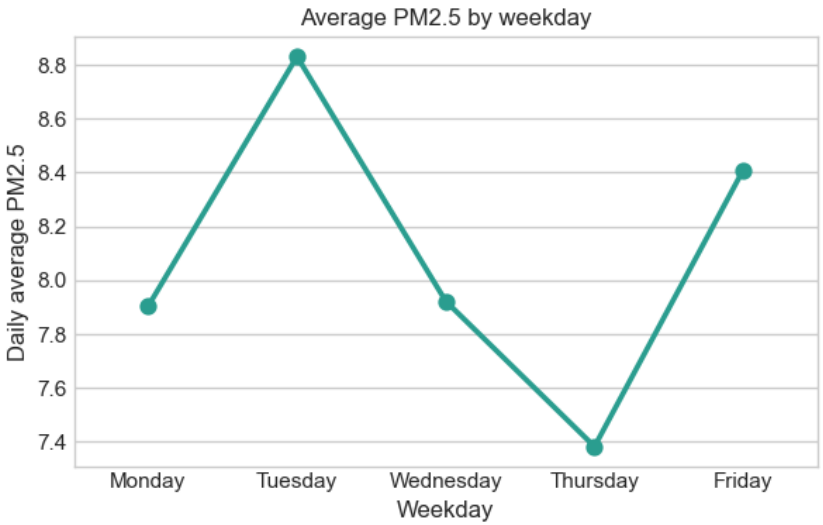


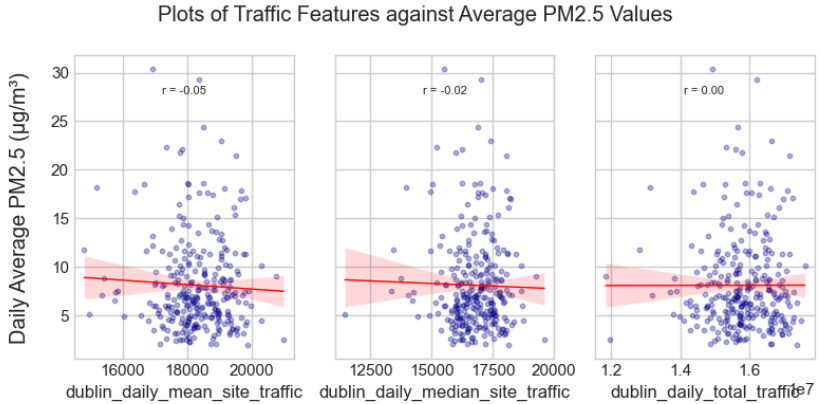


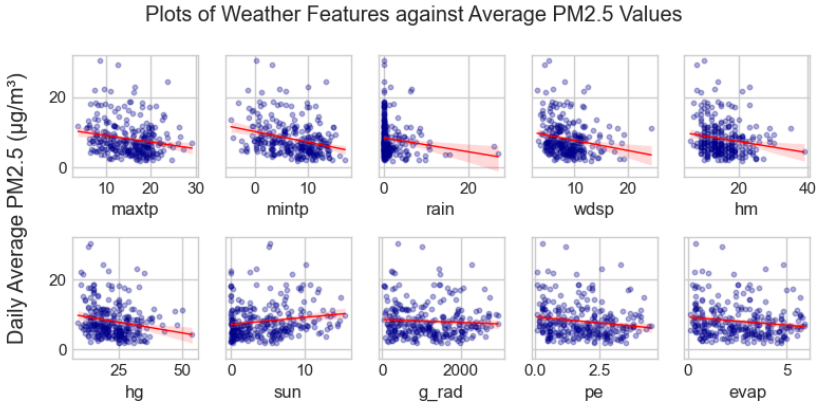
Pre-Processing

- Aggregation and Cleaning:
- Merging:
- Limitations and Considerations:



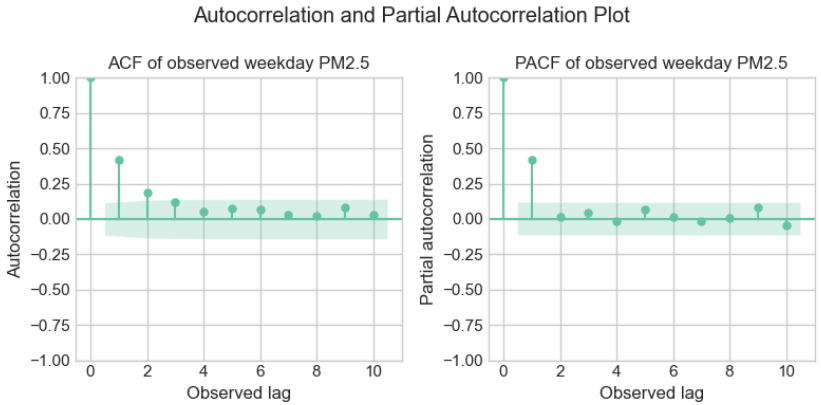


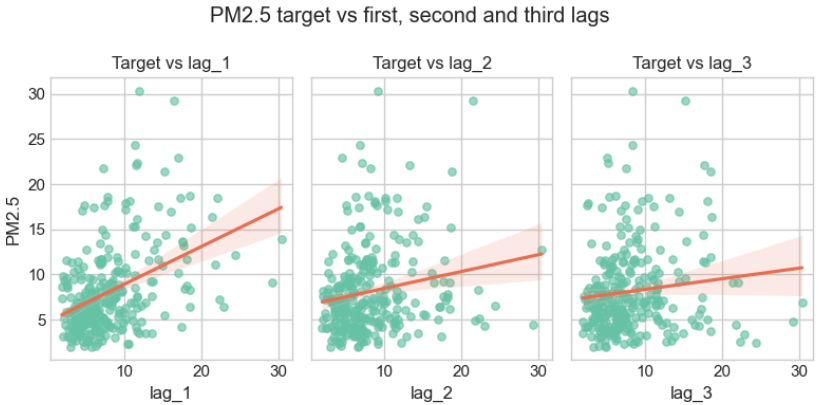




Method of Analysis

- Models
- Feature Selection:
- Baseline:
- Limitations and Considerations:

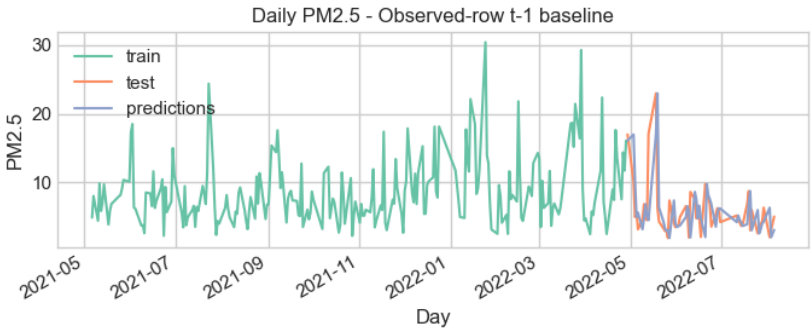


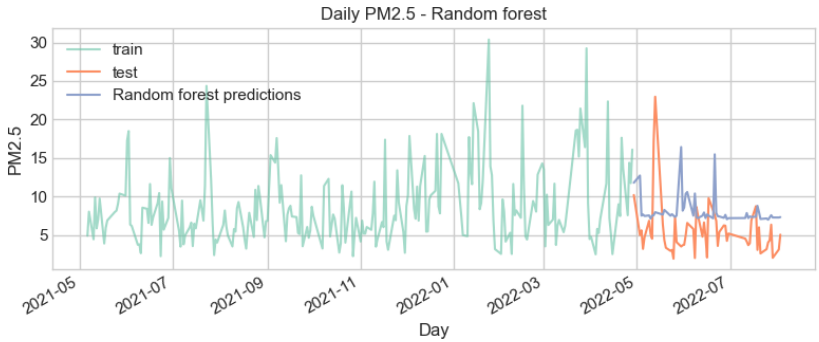


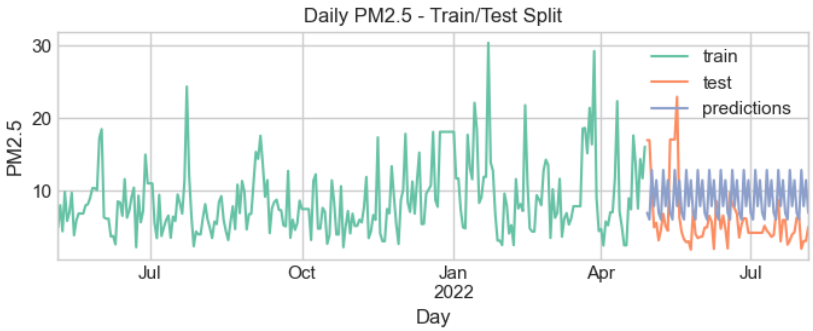
Results

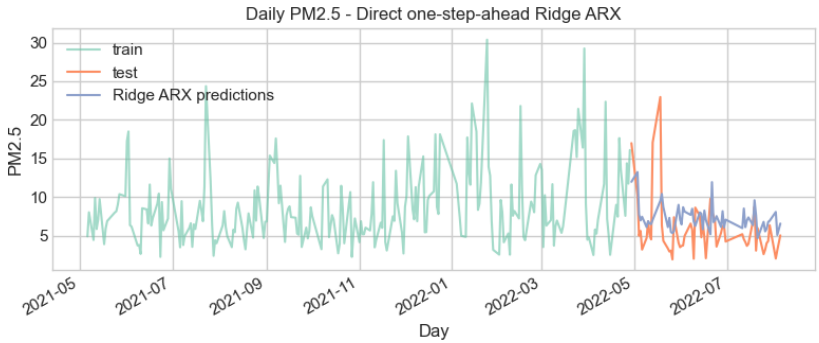
- Graphs / Evaluation Metrics / Feature Importance

Results


Daily PM2.5 - Observed-row t-1 baseline
train
test
predictions
PM2.5
Day


Daily PM2.5 - Random forest
train
test
Random forest predictions
PM2.5
Day


Daily PM2.5 - Train/Test Split
train
test
predictions
PM2.5
Day


Daily PM2.5 - Direct one-step-ahead Ridge ARX
train
test
Ridge ARX predictions
PM2.5
Day

Discussion and recommendation

- Final Model Conclusion
- Benefits of the model
- Limitations and Improvements
- Final Takeaway

Any Questions?

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